Name: ______________________________________ Period: _______ Date: ____________

HW 4.2.0
Angles in a Circle: Degrees and Radians

Review of Lesson 4.2.0

1) What is a radian? Describe the relationship between the radius and circumference of a circle. What are the units/dimensions of a radian?

2) How do you convert from radians to degrees? How do you convert from degrees to radians?

3) What is a co-terminal angle? Give a definition and example.

4) What is a reference angle? Give a definition and example.

Converting Between Radians and Degrees

5) Given the angle in either degree or radian measure, convert it into the other unit of measure.

a) $90°$	b) $225°$	c) $\frac{\pi}{4}$
d) $\frac{7\pi}{4}$	d) $-30°$	e) $-\frac{2\pi}{3}$
f) $150°$	g) $-\frac{11\pi}{6}$	h) $\frac{3\pi}{15}$

Co-terminal and Reference Angles

6) Find an angle between $0°$ and $360°$ that is co-terminal with an angle of…

a) $443°$ b) $-270°$ c) $-1300°$

7) Find an angle between 0 and 2π radians that is co-terminal with an angle of…

a) $\frac{23\pi}{9}$ b) $-\frac{7\pi}{2}$ c) -3π

8) Given the angle θ, determine the quadrant in which the angle terminates and find the measure of θ', the reference angle.

a) $\theta = 275°$ b) $\theta = -579°$

c) $\theta = \frac{13\pi}{6}$ d) $\theta = -\frac{3\pi}{5}$

9) Given the reference angle θ' and quadrant, find the angle θ such that $\theta \in [0, 2\pi)$.

a) $\theta' = \frac{\pi}{4}$; Quadrant III b) $\theta' = \frac{\pi}{3}$; Quadrant II

c) $\theta' = \frac{\pi}{6}$; Quadrant IV d) $\theta' = \frac{2\pi}{15}$; Quadrant III

Trig Values

10) For $\sin\theta = \frac{4}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find the following trig values.

a) $\cos\theta$

b) $\tan\theta$

c) $\sec\theta$

d) $\csc\theta$

e) $\cot\theta$

11) For $\cot\theta = \frac{15}{8}$ and $\pi < \theta < \frac{3\pi}{2}$, find the following trig values.

a) $\sin\theta$

b) $\cos\theta$

c) $\tan\theta$

d) $\sec\theta$

e) $\csc\theta$

Looking Ahead…

12) Review the 45-45-90 and the 30-60-90 triangle rules you learned from geometry.

a) What does each rule say about a right triangle with a height, x?

b) Give the length of all three sides for both types of triangles if the height is 1 unit.

c) Give the length of all three sides for both types of triangles if the hypotenuse is 1 unit.

d) Consider: if the 45-45-90 and 30-60-90 rules are in degrees, then what would be the name of the rule in terms of *radians*?